$$\underset{n\times d}{X}\ \underset{d\times 1}{\theta} = \underset{n\times 1}{y}$$

X → design matrix

y → measurement

$$\begin{bmatrix} X^{(1)} \\ \vdots \\ X^{(n)} \end{bmatrix} \begin{bmatrix} \theta \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Goal: Given X, y, find θ.

Constraint: X $n = d$, X be full rank ← unique sol.

$d \gg n$ ← Underdetermined

$n > d$, X be full rank ← Overdetermined

$$X\theta^* = y$$

Nullspace $\mathcal{N} = \{\Delta \in \mathbb{R}^d \mid X\Delta = 0\}$

$$\theta' = \theta^* + \Delta, \quad \Delta \in \mathcal{N} \rightarrow X\theta' = X(\theta^* + \Delta) = X\theta^* + \underbrace{X\Delta}_{0} = X\theta^* = y$$

Goal: Uniquely determine θ from X & y

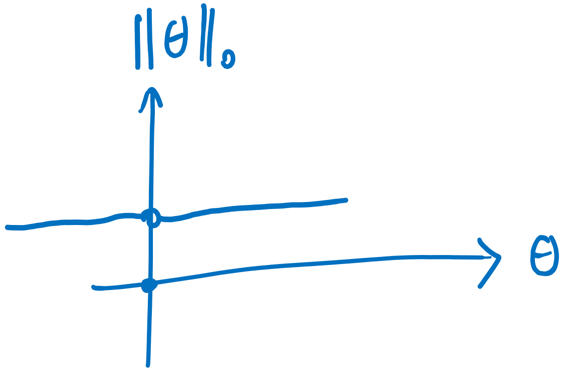
Assumption: θ is sparse $\|\theta\|_0 = s \ll d$.

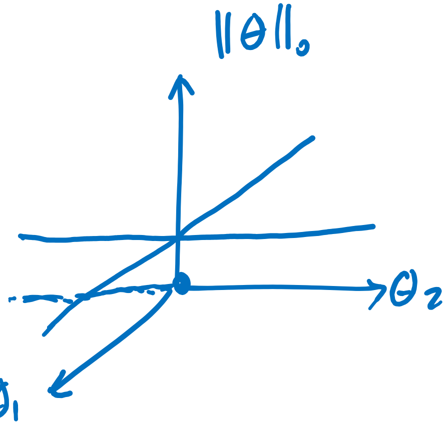
$$\min_\theta \|\theta\|_0 \quad \text{s.t.} \quad X\theta = y$$

$\longrightarrow$ Brute Force

Comb. Search on support of θ. $\binom{d}{s}$.

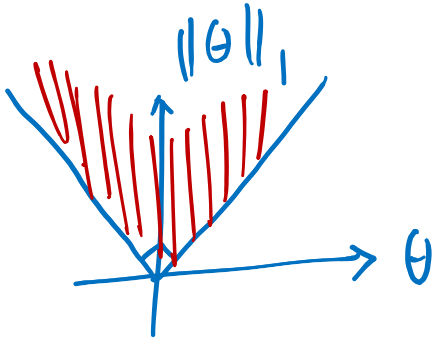
$n \gg s$.





Relax

$$\min_\theta \|\theta\|_1 \quad \text{s.t.} \quad X\theta = y$$



Assumptions on X, such that the sol. to $X\theta = y$ becomes unique, given sparsity of θ.

$$\underbrace{\mathcal{N} + \theta^*}_{} \quad \text{sol.}$$

$$P = \{ \Delta \mid \|\theta^* + \Delta\|_1 \leq \|\theta^*\|_1 \}$$

$$\underbrace{\mathcal{N} \cap P = \{0\}}_{} \longrightarrow \text{then } \theta^* \text{ is the unique sol. of the relaxed opt.}$$

Restricted Nullspace Property RNP

$S \subseteq \{1, \dots, d\}$

$$\mathbb{C}(S) = \{\Delta : \|\Delta_S\|_1 \geq \|\Delta_{S^c}\|_1\}$$

Redef:

$$\mathbb{C}(S) \cap \mathcal{N} = \{0\}$$

Th. These two statements are equivalent:

a) Relaxed Opt has the unique sol. θ^*. ↳ (The one with lowest ℓ^0 norm & satisfy $X\theta^* = y$)

b) $S = supp(\theta^*)$, $\mathbb{C}(S) \cap \mathcal{N} = \{\underline{0}\}$.

indices of θ^* that are non-zero

$\|\theta^*\|_0 = |S| =: s$

Proof: $b \Rightarrow a$

$\cancel{\|\theta^*_S\|_1} = \|\theta^*\|_1 \geq \|\hat{\theta}\|_1$ → sol. of opt. relaxed 0

$\Delta := \hat{\theta} - \theta^*$

$\hat{\theta}_S = \theta^*_S + \Delta_S$

$\hat{\theta}_{S^c} = \underbrace{\theta^*_{S^c}}_{\underline{0}} + \Delta_{S^c}$

$\|a+b\|_1 \geq \|a\|_1 - \|b\|_1$

$= \|\underbrace{\hat{\theta}_{S^c}}_{\Delta_{S^c}}\|_1 + \|\underbrace{\hat{\theta}_S}_{\theta^*_S + \Delta_S}\|_1$

$\Rightarrow \|\Delta_S\|_1 \geq \|\Delta_{S^c}\|_1 \qquad \geq \|\Delta_{S^c}\|_1 + \cancel{\|\theta^*_S\|} - \|\Delta_S\|_1$

Proof by Contradiction : $\Delta = \hat{\theta} - \theta^* \neq 0$

① $\Delta = \hat{\theta} - \theta^* \neq \underline{0}$

② $X\Delta = \underline{0}$ b/c $X(\hat{\theta} - \theta^*) = \underbrace{X\hat{\theta}}_{y} - \underbrace{X\theta^*}_{y} = \underline{0}$

③ $\|\Delta_S\|_1 \geq \|\Delta_{S^c}\|_1$

$\Delta \in \mathcal{N} \cap \mathcal{C}(S) \leftarrow \{$ ②, ③

$\Delta \neq 0$

$\Rightarrow$ RNP is volated ⨳

$\underset{\smile}{a} \overset{?}{\Rightarrow} b$

$\theta^* \in \mathcal{N}(X)$

$$\min_{\beta} \|\beta\|_1 \quad \text{s.t.} \quad X\beta = X\begin{bmatrix} \theta_S^* \\ 0 \end{bmatrix}.$$

$$\hat{\beta} = \begin{bmatrix} \theta_S^* \\ 0 \end{bmatrix}$$

$$X\theta^* = \underline{0} \Rightarrow X\left(\begin{bmatrix} \theta_S^* \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \theta_{S^c}^* \end{bmatrix}\right) = 0 \Rightarrow X\begin{bmatrix} \theta_S^* \\ 0 \end{bmatrix} = X\begin{bmatrix} 0 \\ \underset{\smile}{-\theta_{S^c}^*} \end{bmatrix}$$

for the opt. to have unique sol.

$$\left\| \begin{bmatrix} \theta^*_S \\ 0 \end{bmatrix} \right\|_1 < \left\| \begin{bmatrix} 0 \\ -\theta^*_{S^c} \end{bmatrix} \right\|_1$$

$$(*) \quad \|\theta^*_S\|_1 < \|\theta^*_{S^c}\|_1$$

Proof by Contradiction, Assume there exist some $\theta^* \in \mathcal{N}$ $\|\theta^*_S\|_1 \geq \|\theta^*_{S^c}\|_1$. Replace this θ^* in the above argument.

For this, $(*)$ is deduced $\Rightarrow$ $\times$

$\underline{\text{Cond}}\,(X) \Rightarrow \text{RNP}(X) \Rightarrow \hat{\theta} = \theta^*$

Th. $$\delta_{pw}(X) := \max_{i,j} \left| \left| \frac{\langle X_i, X_j \rangle}{n} \right| - \mathbb{I}(i=j) \right|$$

$$X = \begin{bmatrix} X_1 & \cdots & X_d \end{bmatrix}$$

If $\delta_{pw}(X) \leq 1/3s \Rightarrow \text{RNP}$